Light Field Depth Estimation for Non-Lambertian Objects via Adaptive Cross Operator

Zhenglong Cui, Hao Sheng, *Member, IEEE*, Da Yang, Sizhe Wang, Rongshan Chen, and Wei Ke

***Abstract*— Light field (LF) depth estimation is a crucial basis for LF-related applications. Most existing methods are based on the Lambertian assumption and cannot deal with non-Lambertian surfaces represented by transparent objects and mirrors. In this paper, we propose a novel Adaptive-Cross-Operator-based(ACO) depth estimation algorithm for non-Lambertian LF. By analyzing the imaging characteristics of non-Lambertian regions, it is found that the difficulty of depth estimation lies in the photo inconsistency of the center view. Combining with the two-branch structure, we propose ACO with an inter-branch cooperation strategy to adaptively separate depth information with different reflectance coefficients. We discover that the bimodal distribution feature of the operator filtering results can assist in the separation of multi-layer scene information. The first detection branch filters the EPI and implicitly records the severity of multi-layer scene aliasing. According to the identification of bimodal distribution features, the non-Lambertian regions are marked out and the depth of the foreground is estimated. The second branch receives guidance from the first to dynamically adjust the inner weight and infer the background's depth after weakening the interference from the foreground. Finally, the depth information separation of multi-layer scenes is achieved by extracting the unique X-shaped linear structure. Without the reflection coefficients of the non-Lambertian object, the proposed method can produce high-quality depth estimation under the transparency of 90% to 20%. Experimental results show that the proposed ACO outperforms state-of-the-art LF depth estimation methods in terms of accuracy and robustness.**

***Index Terms*— Light field, non-Lambertian, depth estimation, non-Lambertian region detection, adaptive cross operator.**

I. Introduction

CONVENTIONAL photography captures the cumulative intensity of light rays on the 2D imaging plane and loses the direction information of light inevitably. In comparison, light field (LF) cameras record both spatial and angular light ray information in a single shot. This unique 4D dense sampling characteristic provides a new method for 3D scene reconstruction. As a fundamental task in LF image processing, LF depth estimation has been widely explored in the last few years. Most works like [1], [2], [3], and [4] are based on a common Lambertian assumption that one scene point is observed as the same intensity in different views. Take it a step further, when fixing an LF image's spatial coordinates and refocusing pixels from the correct depth, the obtained angular patch has photo consistency [5]. However, the Lambertian assumption is broken where transparent objects and specular surfaces are present, which results in erroneous depth estimates.

For Lambertian scenes, photo consistency is the core theoretical basis of depth estimation algorithms. Specifically, correspondence and defocus are two typical costs that are used to measure photo consistency. Depth estimation algorithms refocus the angular patches from different candidate depth labels to compute cost volumes, where the lowest cost and the highest consistency indicate the selection of the optimal depth. These Lambertian-assumption-based depth estimation algorithms perform well in most scenes. On the contrary, the correspondence and defocus cost generate very blurry depth estimation in non-Lambertian areas because pixels in the refocused angular patch may come from a reflector. This causes existing methods to underestimate their depth or even wipe out the structures. Accordingly, Lambertian-assumption-based algorithms generate many artifacts in transparent or mirror areas, as shown in Fig. 1. Nevertheless, there are currently limited means of processing for non-Lambertian region depth estimation. It is even proposed in [7] that 'We exclude some training data that contains reflection and refraction regions, . . .which result in incorrect correspondences. **Reflection and refraction regions were manually masked out**.' Although [8], [9], and [10] have proposed methods concentrating on non-Lambertian objects, their performance is limited by noise and reflectance.

In this paper, we innovatively solve the problem of depth estimation for scenes with different transparency and reflectance. We find the X-shaped structure on the EPI as the key breakthrough. As shown in Fig. 2, the EPI of the transparent and specular regions exhibit a unique identical X-shaped structure, while the Lambertian region is simply a one-way straight-line structure. This difference is due to the

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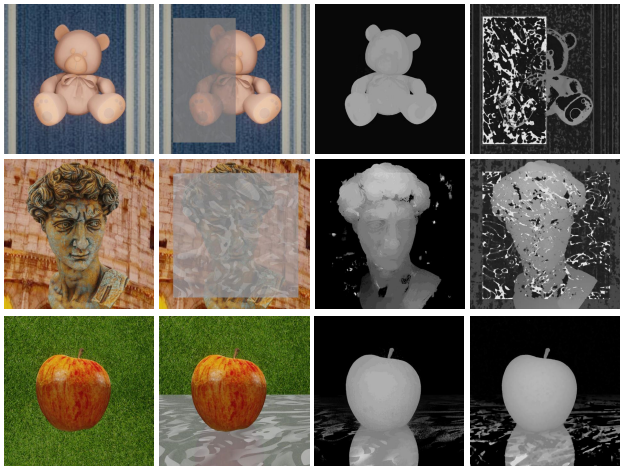


Fig. 1. Depth estimation for scenes with transparent objects or mirrors. ACO proposed in this paper infers the correct depth of objects in non-Lambertian regions. The first column contains the Lambertian object appearance. The second column contains the central view image of the non-Lambertian scene. The third column contains the depth estimation result of our proposed method. The fourth column contains the depth estimation result of OAVC [6] as a comparison.

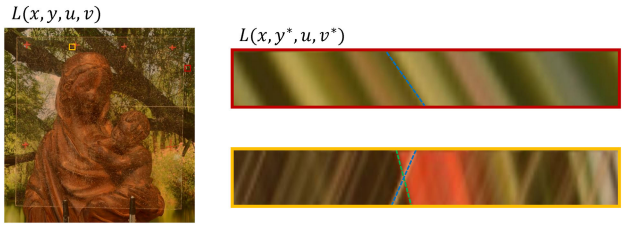


Fig. 2. EPI linear structure difference between Lambertian and transparent regions. In Lambertian region, the lines are arranged in sequence. However, in transparent region, due to the overlay of foreground and background information, an X-shaped linear structure is formed on EPIs.

superposition of multi-layer scene information with different depths in non-Lambertian regions. And we try to leverage such a special structure to separate the depth information of Lambertian and non-Lambertian objects in the case of unknown reflection coefficients.

We designed the X-shaped Adaptive Cross Operator(ACO) consisting of two branches. The separation of aliased scene information is achieved by the cooperation of the two branches. Our initial depth map estimation has the following two main steps: detect the aliasing severity of non-Lambertian areas, and estimate the background depth with the dynamical weight. Specifically, the first branch filters EPI, whose filtering results implicitly include the non-Lambertian object distribution, reflection coefficients, and noise conditions. This information is used not only to mark out non-Lambert regions and estimate the depth of foreground objects but also to infer reflection coefficients and guide the second branch. Then the second branch changes its intra-branch weights to weaken the interference of the foreground and extracts the depth information of the background. This double-branch-cooperation idea can effectively handle non-Lambertian scenarios with unknown reflection coefficients. Subsequently, we experimentally demonstrate the effectiveness and robustness of the algorithm from quantitative, by the comparison with the state-of-the-art methods on the public 4D Light-Field Dataset of HCI [11]. In addition, in order to prove the robustness of the algorithms to different transparency and reflectance, we generated synthetic data and designed experiments with control variables to quantitatively prove it. The experimental results show that the proposed algorithm obtains accurate estimation results in the transparency range of 90% to 20%.

The main contributions of this paper include 4 parts,

- Misinformation arising from superimposed central view is found to be the main cause of non-Lambertian depth estimation errors. Inconsistency caused by the central view invalidates the algorithms based on Lambertian assumption.
- ACO is designed to match the X-shaped structure on EPI and an adaptive module enables ACO dynamically adjust weights to different reflection coefficients.
- The bimodal distribution feature of ACO filtering results is revealed to be applicable for non-Lambertian region detection and multi-layer scene information separation.
- An ACO-based depth estimation algorithm is proposed for LF, especially in non-Lambertian scenes. The experimental results show that the algorithm performs well by comparison with the state-of-the-art methods on the public 4D Light-Field Dataset of HCI [11]. And experiments with controlled variables are conducted to prove that ACO is robust to different transparency and reflectance.

We briefly summarize the research status in Sec. II. In Sec. III, We analyze the imaging characteristics of non-Lambertian regions and introduce the well-engineered ACO to match the X-shaped linear structure on EPI. Based on this, we propose a depth estimation algorithm for non-Lambertian scenes in Sec. IV. Extensive experiments in Sec. V verifies the effectiveness of the proposed theory.

II. Related Work

Compared with conventional cameras, LF cameras can capture the light intensity of an object in a higher angular dimension. Relying on such property, LF is widely used in saliency detection [12], [13], [14], super-resolution [15], [16], [17], [18], [19], semantic segmentation [20], [21], light field fusion [22] and material recognition [23], [24]. For LF depth estimation for non-Lambertian scenes, we briefly introduce the research status from three aspects: LF depth estimation, non-Lambertian scene reconstruction, and LF depth estimation for non-Lambertian scenes.

A. LF Depth Estimation

As a fundamental task in LF image processing, much research has been made to improve the accuracy of depth estimation. Yu et al. [25] propose a line-assisted graph-cut algorithm. They encode 3D line constraints in stereo matching to improve light field triangulation. Chen et al. [26] develop a bilateral consistency metric to handle significant occlusion. The basis of their method is matching the same scene point in different sub-aperture. Wang et al. [27] propose the edge of occlusion and the line of occlusion have the same orientation. The error caused by occlusion can be effectively reduced by modeling such directional consistency. Shin et al. [7] propose a fully-convolutional neural network using EPI geometry. Their framework is so classical that it provides a basis for subsequent

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research. Sheng et al. [28] develop a strategy to make full use of multi-orientation EPIs. Sheng et al. [28] analyze the geometric relationship between angular sampling images and reference sub-aperture images, and then prove the occlusion boundary similarity. Zhang et al. [3] design a two-stage light field depth estimation method via graph spectral analysis. Huang et al. [29] design a lightweight disparity estimation model with physical-based multi-disparity-scale cost volume aggregation for fast disparity estimation. Wang et al. [30] propose enhanced Spinning Parallelogram Operator utilizing color constraint and histogram integration to handle complex occlusions. Chen et al. [2] propose an attention-based multi-level fusion network to selecte features of views with less occlusions and richer textures by the attention mechanism. Wang et al. [31] develop a DistgDisp using disentangling mechanism for disparity estimation. The basic assumption of the above methods is similar. There is only diffuse reflection in the scene, and the same object looks the same from different views. These methods are effective in solving the occlusion problem in Lambertian scenes. However, they tend to cause significant errors when non-Lambertian objects are present.

B. Non-Lambertian Scene Reconstruction

Much previous research has been developed in non-Lambertian scene reconstruction under controlled lighting conditions [32], [33], but still difficult to generalize to images under unknown illumination. Some learning-based methods [34], [35] have been proposed to detect transparent regions for normal 2D images. Fritz et al. [36] model transparent locals patch appearance using an additive model of latent factors and aim at recognizing transparent glass. Wang et al. [37], [38] exploit RGBD cameras and introduce depth information to localize glass objects. For light field, Oo et al. [39] separate out reflection, and Tingtian et al. [40] remove reflection but only for solid color transparent objects. Li et al. [41] propose a deep inverse rendering framework for indoor scenes. They reconstruct a complete scene including shape, spatially-varying lighting, and non-Lambertian surface reflectance from a single RGB image. Recently, the reconstruction method of non-Lambertian scenes using light field has also been proposed. Moreschini et al. [42] extend the shearlet-based reconstruction algorithm to enable LF reconstruction for scenes composed of surfaces with non-Lambertian properties. Zhou et al. [43] design a concentric multi-spectral LF to recover the shape and reflectance of surfaces with arbitrary material.

C. LF Depth Estimation for Non-Lambertian Scenes

Recently, some works are proposed to solve depth estimation for non-Lambertian objects. Tao et al. [44] develop an iterative approach to remove the specular component and improve depth estimation by estimating light source colors. Then they present line consistency with a method to estimate the light source color and separate specularity in [45]. They classify the pixels as either Lambertian or specular, which cannot handle general glossy surfaces. To obtain the shape of glossy objects, Ting et al. [46] derive a spatially-varying-BRDF-invariant theory for recovering the 3D shape and the reflectance. The key theoretical insight is an analysis of diffuse plus single-lobe SVBRDFs under a light-field setup.

In order to solve the depth estimation problem of multi-layer scenes, Wanner et al. [8] develop second-order structure tensors to model reflecting and transparent surfaces. The two disparities can be calculated by the eigenvalues of structure tensors. Based on this double-orientation model, the authors separate the depth information of Lambertian and non-Lambertian objects. Johannsen et al. [9] propose using sparse decomposition to leverage the depth-orientation relationship on epipolar plane images. The sparse-coding-based method learns the structure of the central view and constructs an LF dictionary. Then this dictionary is then used to encode a sparse representation of the LF. Alperovich et al. [10] present a fully convolutional autoencoder for LF, which reduces the structure of LF to a low-dimensional representation. This method can effectively separate the specular region from the diffuse scenes. The experimental results prove the effectiveness of these algorithms in non-Lambertian region depth estimation.

Nevertheless, several drawbacks of existing non-Lambertian depth estimation methods limit their performance. The double-orientation models [8] lack credible theoretical support. The author acknowledges in the article that "Unfortunately, we do not know yet of a reliable mathematical measure which tells us whether the two-layer model is valid or not." Besides, both double-orientation models [8] and LF sparse-coding [9] are sensitive to texture noise. When large patches of pure color appear on non-Lambertian surfaces, these methods obtain large areas of artifacts. As a local feature, the structure tensor cannot deal with noise beyond the receptive field. For sparse coding, solid-color patches pollute the quality of the dictionary during the iteration and lower the expression ability of the dictionary. In addition, the transparency of non-Lambertian objects and the disparity range of LF images also affect the effectiveness of the algorithm, which is proved by experiments in Sec. V. And for deep encoder-decoder network [10], it mainly deals with the problem of highlight, rather than the multi-layer depth problem caused by refraction and reflection. Therefore, the problem of estimating the depth of non-Lambertian scenes remains challenging.

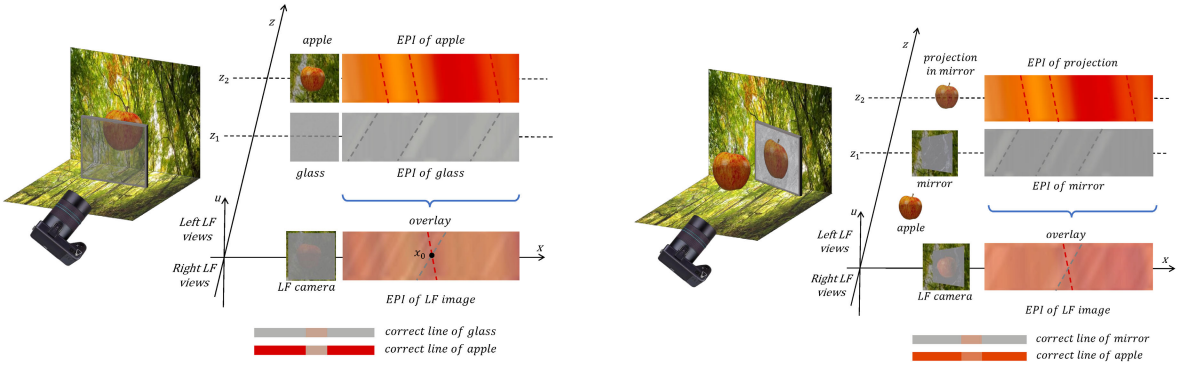
III. Adaptive Cross Operator

In this section, we contribute an analysis of the unique X-shaped linear structure in the mirror and transparent region on EPI. We find that the photo inconsistency on the X-shaped structure and the superposition of the central view are the main reasons for the inaccurate depth estimation of multi-layer scenes. Therefore, we design an X-shaped ACO to match this special structure.

A. Photo Consistency of Lambertian Scenes

In the beginning, we explain the used LF two-plane-parameterization method. Denote by $L(x, y, u, v) \in \mathbb{R}^{X\times Y\times U\times V}$ a 4D light field image, where x, y are spatial coordinate and u, v are angular coordinates. If the two coordinates

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(a) LF simulation of transparent glass.

(b) LF simulation of mirror.

Fig. 3. LF simulation of transparent and specular region. By setting Lambertian object apple, transparent glass and mirror separately, the EPI generated by the corresponding object itself can be obtained. Thus it can be proved that the EPI of transparent and mirror area is the superposition of two layers of scene information.

(y^*, v^*) is fixed, an EPI $L_{(y^*,v^*)}(x,u)$ is obtained. An EPI can be interpreted as a horizontal or vertical slice through a horizontal or vertical stack of views. Due to the different disparity between adjacent views, the overlaid straight lines with different slopes form a rich linear structure on EPI. Disparity d has a reciprocal relationship with slope $tan\theta$

$$d = \frac{1}{tan\theta} \tag{1}$$

In the follow-up analysis, we estimate discrete depth parameters $o \in O = 1, 2, \ldots, o_{max}$, where o_{max} is the depth resolution. The relationship between depth label o and disparity d is defined as

$$d(o) = d_{min} + \frac{d_{max} - d_{min}}{o_{max} - 1}(o - 1), \tag{2}$$

where d_{min} and d_{max} are the minimum and maximum disparities between adjacent views. And $1/d_{max}$ and $1/d_{min}$ are the minimum and maximum slopes of the lines on EPIs.

For Lambertian scenes depth estimation, matching the lines on EPI determines the accuracy of the results. Specifically, refocus on different candidate disparities to build a cost volume and collect the refocused pixels. Then calculate the variance of the pixels on the obtained line to measure the photo consistency. To be specific, we consider a point x_0 of the central view ($u = 0$) on the EPI in Fig. 3. For a given slope $tan\theta$ for the depth o, the angular variance can be calculated by

$$\sigma_o(x_0)^2 = \frac{1}{U-1}\sum_u [L(x_0 + \frac{u}{tan\theta}, u) - \overline{L}_o(x_0)]^2 \tag{3}$$

where $\overline{L}_o(x_0)$ is the angular mean along the line for o:

$$\overline{L}_o(x_0) = \frac{1}{U}\sum_u [L(x_0 + \frac{u}{tan\theta}, u)] \tag{4}$$

The angular variance $\sigma_o(x_0)^2$ described the correspondence of pixels on a candidate refocused line. Based on the Lambertian assumption, the highest consistency is obtained by refocusing the correct depth. The depth that produces the least angular variance is considered the estimated result $o(x_0)$,

$$o(x_0) = \arg\min \sigma_o(x_0) \tag{5}$$

However, this kind of method is not suitable to deal with the X-shaped structure in non-Lambertian scenes. This difference in structure is shown in Fig. 2. For depth estimation of non-Lambertian scenes, the main difficulty is to infer the depth of the reflector and the reflected object separately. Ideally, matching the two branches of the X-shaped structure can realize the separation of the two-layer scene of foreground and background. However, the limited angular resolution and unknown reflectance parameters make accurate matching challenging.

B. Photo Inconsistency of Non-Lambertian Scenes

We present an analysis of X-shaped structures' formation in non-Lambertian regions. Let's first consider transparent scenes. In Fig. 3(a), suppose two objects are placed at different distances from the LF camera. The nearer object is a transparent gray glass whose distance is denoted as z_1 completely occludes the further red apple with the distance denoted as z_2. The texture on the glass forms lines on the EPI of the foreground, which is displayed on the right side of the glass. The pattern on the apple also forms overlaid straight-line structure on the corresponding EPI, as shown on the right of the apple. The LF camera is placed at $z = 0$ and the EPI of the LF image is visualized on the right.

1) X-Shaped Linear Structure of the Transparent Area: Starting with the analysis of the apple in Fig. 3(a), as a common Lambertian object, its EPI shows a standard straight-line structure. The red lines on EPI are arranged in order and their slopes correspond to depth z_2. Similarly, when only considering the glass, its texture forms gray straight lines whose slopes correspond to the depth z_1. For the LF camera at $z = 0$, the light captured by the lens can ideally be divided into two parts, diffuse light from the glass surface and light from the apple refracted through the glass. Information from the two-layer scene is overlapped on EPI and forms an X-shaped line structure. And the two branches of the X-shaped structure correspond to the depth information of z_1 and z_2 respectively. Therefore, as long as the two branches are correctly separated, the depth information of the two layers can be estimated.

2) X-Shaped Linear Structure of the Mirror Area: Then we consider scenes with specular reflection. In Fig. 3(b), a gray

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smooth marble like a mirror is placed at depth z_1, and its diffuse reflection forms gray lines on EPI as shown. A red apple is placed in front of the marble, and the projection of this apple is reflected in the marble at depth z_2. The diffuse light from the apple is reflected by the smooth marble and captured by the camera. This is equivalent to the apple's projection observed through the mirror surface. Thus, mirrors are similar to transparent objects on EPI. Information of Lambertian and non-Lambertian objects are superimposed on the imaging plane and an X-shaped structure is formed due to the difference in depth. Accordingly, this unique X-shaped structure in the refraction and specular reflection regions makes the problem of multi-layer scene depth estimation unified. The problem of depth estimation for transparent and mirror regions can be transformed into matching and separation of X shape's two branches.

As mentioned above, for Lanbertian scenes, the process of depth correspondence matching can be considered as calculating pixel intensity variances along different candidate lines on EPI [47]. Depth correspondence matching has good performance for most Lambertian regions. But this angular consistency breaks down in the non-Lanbertian region. As shown at the bottom of Fig. 3, the correct refocused line of the apple loses its original consistency. The pixels in the center view are occluded and overlapped with the colors of non-Lambertian objects. This results in a large angular variance due to the inconsistency between the center point and the rest of the pixels. Thus, even though the apple has a standard diffuse surface, correspondence matching is not available for inferring the depth of apple.

So we can summarize the main problem caused by X-shaped structure: **Photo inconsistency in the X-shaped linear structure.** The pixel at the center view is usually seen as a correct reference point, i.e. the pixels on the correctly matched line should be consistent with the point of the center view. However, the color and depth information of Lambertian and non-Lambertian objects overlap in the center view. The center point is a different color from the correctly matched line in an X-shaped structure. Therefore, the Lambertian assumption and high angular consistency can no longer be used to judge whether the result of depth estimation is correct. Especially in the case of unknown transparency and reflectance, the component separation of multi-layer information at the central view becomes difficult to achieve. Hence, only considering the consistency of pixels on the line can not accurately identify the X-shaped structure. We decide to exploit the surrounding pixel information of the X-shaped structure for matching.

C. The Design of ACO

For the X-shaped linear structure of non-Lambertian regions, common stereo-matching-based methods have been unable to effectively solve the challenge caused by photo inconsistency. Thus, we design ACO of a similar X-shaped structure for two-layer depth estimation.

As shown in Fig. 4, ACO consists of two parallelogram branches of different widths. The centers of the two branches coincide. Inside each branch is a center line of variable slope. When the centerline rotates around the center point, the corresponding ACO branch rotates with it. Each centerline divides the branch into left and right parts and the color difference between the two parts is compared during the rotation process. Ideally, the two branches of ACO obtain the maximum response value at two branches of an X-shaped structure respectively.

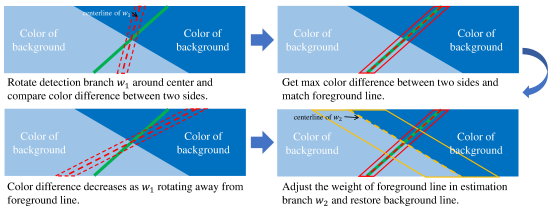


Fig. 4. ACO matches non-Lambertian lines by rotating the first detection branch. The result leads the estimation branch to infer the straight line of Lambertian objects. In this way, the depth information separation of the two-layer scene is realized.

Specifically, take the center of ACO as origin $(0, 0)$ and (i, j) as the relative coordinate, the corresponding pixels in ACO for depth label o are weighted as follows:

$$w_1(i, j, o_1) = c * (i - j * d(o_1)) * e^{\frac{-(i-j*d(o_1))^2}{2{\alpha_1}^2}} \tag{6}$$

$$w_2(i, j, o_1, o_2) = \begin{cases} \beta_t * c * (i - j * d(o_2)) * e^{\frac{-(i-j*d(o_2))^2}{2{\alpha_2}^2}}, \\ \text{if } |w_1(i, j, o_2)| \geq 0.01; \\ c * (i - j * d(o_2)) * e^{\frac{-(i-j*d(o_2))^2}{2{\alpha_2}^2}}, \\ \text{if } |w_1(i, j, o_2)| < 0.01; \end{cases} \tag{7}$$

where w_1 is the first detection branch and w_2 is the second estimation branch. o_1 and o_2 are depth parameters for the two branches. α_1 and α_2 are scale parameters that control the width and sensitivity of each branch, $\alpha_2 = 1.2 * \alpha_1$ by default. β_t is a weakening parameter to weaken the effect of foregrounds. c is a constant to control the total weight. The optimal parameter selection is introduced in the experiments.

ACO convolves on the color histogram h of EPIs and calculates the color distance. The color histogram is divided into 64 bins and takes the sum of the convolution in each channel. The use of color histogram distance instead of RGB value is to avoid excessive color difference of single channel resulting in local extreme value. The response value function f_1 and f_2 of two branches can be obtained as follow:

$$f_1(i, j, o_1) = w_1(i, j, o_1) * h, \tag{8}$$

$$f_2(i, j, o_1, o_2) = w_2(i, j, o_1, o_2) * h. \tag{9}$$

For depth parameters $o_i (i = 1, 2)$, each branch takes the line with slope $1/d(o_i)$ through the origin as the centerline. The weights radiate from the centerline to the two sides. The two sides' weights have different positives and negatives. Therefore, the convolution result is the color difference between sides. For the basic working principle of ACO, as shown in Fig. 4. As the depth parameter changes, the branch rotates around the center point. When the centerline coincides with the line to be estimated, the response value of ACO reaches the maximum value.

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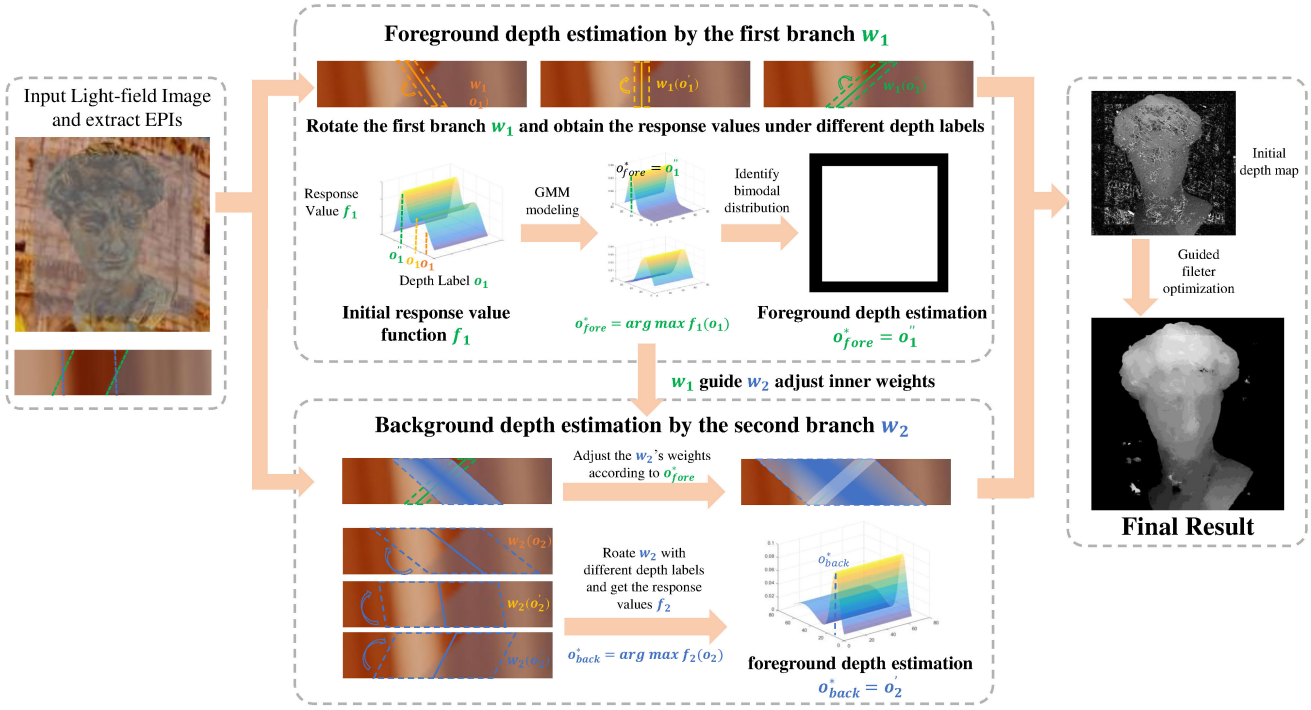


Fig. 5. Schematic diagram of the proposed ACO-based depth estimation algorithm.

Based on the analysis of the X-shaped linear structure, it is found that the difficulty of depth estimation for non-Lambertian regions lies in the superposition of multi-layer scene information at the center view and the resulting photo inconsistency of matching lines at the correct depth. Therefore, we design ACO to remove the dependence on the central view and intentionally separate the aliased scene information. Next, we describe in detail how the cooperation of these two branches can be used to detect transparent regions and estimate foreground and background disparity. They are used to solve the inadaptability of conventional matching models to non-Lambertian scenes.

IV. ACO-Based Depth Estimation for Non-Lambertian Scene

In the previous section, we described how to construct ACO and leverage it to match X-shaped linear structures in non-Lambertian regions on EPI. Next, we present how this can be used to infer the depth of non-Lambertian scenes.

A. Problem Setting

We take transparent objects as representatives for the following discussion. Let $P_l \in R^3$ be a Lambertian object and $P_t \in R^3$ be a transparent object in front of P_l. The P_l and P_t's projections at the center view overlap. In transparent region, as shown in Fig. 2 and Fig. 3, the slope of two straight lines are $1/d_{o_{fore}}$ and $1/d_{o_{back}}$, where $d_{o_{fore}}$ and $d_{o_{back}}$ are the disparity of P_l and P_t. The problem of depth estimation is as follows: Given a point (x, y) on the central view, determine if it is in a non-Lambertian region. If in a non-Lambertian region, the depth information of the foreground transparent object and the background Lambertian object should be estimated. Otherwise, the depth information of the Lambertian object should be inferred.

B. ACO-Based Depth Estimation

The framework of the algorithm is shown in Fig. 5. The ACO-based algorithm can be divided into four main steps: input LF, transparent region detection and foreground depth estimation, background depth estimation, and optimization.

The first detection branch w_1 is used to detect transparent regions. When o_1 changes, the centerline of w_1 rotates. For Lambertian regions, ideally, the color difference between the two sides gradually increases as w_1 gets closer to the straight line. And it obtains maximum response value when the centerline coincides with the line to be estimated. However, it gets complicated when dealing with transparent regions. In transparent regions, due to the X-shaped linear structure on EPI, the response value function f_1 shows a bimodal distribution. However, because of the overlap of foreground and background information, the two peaks may not be distinguishable clearly. Thus, Gaussian Mixture Model(GMM) is used to model the distribution of f_1. Firstly, f_1 is transformed to be a probability distribution function P:

$$P(o_1) = \frac{f_1(o_1)}{\sum_o f_1(o_1)} \tag{10}$$

Use GMM of two sub-distributions to fit P, then the fitted distribution is

$$P'(o_1) = \sum_{k=1}^{2} p_k \varphi(o_1|\lambda = \lambda_k) \tag{11}$$

where $p_k = P(\lambda = \lambda_k)$ is mixing coefficient of the kth sub-distribution. $p_k \geq 0$ and $\sum p_k = 1$. $\varphi(\emptyset_1|\lambda = \lambda_k)$ is

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probability density function of the kth sub-distribution. λ_k is parameter of the kth Gaussian sub-distribution, including mean value μ_k and variance ${\sigma_k}^2$. Exploiting GMM, We have achieved the separation of the two peaks of f_1. Then, the boolean value B_tra that indicates whether a pixel belongs to transparent regions is judged by the rate between two peaks,

$$B_{tra} = bool(min(\frac{\frac{p_1}{\sigma_1}}{\frac{p_2}{\sigma_2}}, \frac{\frac{p_2}{\sigma_2}}{\frac{p_1}{\sigma_1}}) > t_{tra}) \tag{12}$$

where t_{tra} is a threshold to control the sensitivity of transparent region detection.

After obtaining the detection result of transparent regions, the depth estimation of Lambertian regions and non-Lambertian regions can be processed differently. For Lambertian regions, the corresponding line can then be determined from the response value function f_1. The depth estimation result o^* can be obtained as follow:

$$o^* = \arg\max_{o_1} f_1(o_1) \tag{13}$$

For transparent regions, there are two peaks in the response value function f_1 after being fitted by GMM. Due to the transparent object acting as an occluder for the background object, it takes a stronger dominant position in terms of appearance. Therefore, the depth corresponding to the smaller depth label of the two peaks can be chosen as the result o^*_{fore} of the foreground. However, for the background, we cannot simply choose another peak's depth label as the result. Because the straight lines of the background object are interrupted by the foreground and the color information is influenced by the foreground. Thus, the result of the detection branch is used to guide weakening foreground interference. Then the second estimation branch w_2 is needed. In the w_2, the weights that overlap with the first branch are redistributed. The background depth can therefore be obtained by selecting the maximum value of f_2:

$$o^*_{back} = \arg\max_{o_2} f_2(o^*_{fore}, o_2) \tag{14}$$

The final result is optimized using guiding filtering to preserve edges.

C. Adaptive Model Against Noise

Due to the parameters of the ACO being static, the compatibility with different transparency and reflectance needs to be improved. Thus, we design an adaptive module to improve the robustness of ACO.

1) Sensitivity Adapts to Noise: After detection branch filtering, not only the distribution of transparent regions can be obtained, but the severity of the noise is implicitly recorded. If this noise detection information can be used to dynamically adjust the sensitivity of the estimation branch, the effect of noise on depth reduction can be effectively avoided. The scale parameter α_2 can be adjusted by following:

$$\alpha_2 = max(p_1/p_2, p_2/p_1) * \alpha_1 \tag{15}$$

When the ratio of two sub-distributions p_1 and p_2 differs significantly, it infers the information of the background objects is occluded heavily. In this case, the estimation branch needs to be robust and take into account the information of context in order to accurately restore the background depth. Thus, the width of the estimation branch is increased, making it insensitive to occlusion.

2) Estimation Adapts to Transparency: In the original design, regardless of the difference between the slopes of centerlines, the weights of overlapping parts are weakened to the same extent to reduce the effect of foreground on background. The advantage of this is that once the depth of the foreground has been determined, the background can be restored nearly without the interference of foreground occlusion. However, this also creates a drawback for ACO. When the depths of foreground and background are similar, such a strategy can cause the original information of background to be weakened as well. The depth variation of background in estimated results will not be obvious and may lead to a large area of estimation errors. Therefore, we propose to weaken the estimation branch at different depths to different degrees. For branches with large differences in depth labels o_1 and o_2, due to the small overlap area, we chose a stronger degree of attenuation to achieve a clear distinction between foreground and background. On the contrary, for branches with a small difference in depth labels, weak attenuation is used in order not to affect the original background information. Therefore, we re-adjust the weight distribution of the estimation branch w_2 on the basis of Equ. 7:

$$w_2(i,j,o_1,o_2) = \begin{cases} \beta_t * (1-(1-\beta_t) * |o_1 - o_2|/o_{max}) \\ *c * (i - j * d(o_2)) * e^{\frac{-(i-j*d(o_2))^2}{2{\alpha_2}^2}}, \\ \text{if } |w_1(i,j,o_2)| \geq 0.01; \\ c * (i - j * d(o_2)) * e^{\frac{-(i-j*d(o_2))^2}{2{\alpha_2}^2}}, \\ \text{if } |w_1(i,j,o_2)| < 0.01; \end{cases} \tag{16}$$

V. Experimental Results and Discussion

In this section, we first introduce the dataset and metrics we used for evaluation. Then both quantitative and qualitative results are reported and compared with the state-of-the-art methods, along with the ablation study, and discussions.

A. Datasets and Metrics

We evaluate the proposed method on the test set "Transparency" in 4D Light-Field Dataset of HCI from [11], which is a public rendered light field images dataset for depth estimation. Reference [11] contains the disparity range for each LF image. To our knowledge, this dataset is considered to be the first and only benchmark light field for multi-layer disparity estimation. However, this dataset lacks data for different reflection parameters. To better compare algorithms performance, we use the software Blender and tools provided by [11] to generate rendered data. "Teddy" is a scene of a teddy bear half obscured by a glass, which is designed to test the continuity of the background's depth estimation result. "David" is a set of images with the same scene, where a head sculpture statue of David is obscured by a glass of varying transparency.

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TABLE I
COMPARISON ON MSE AND BADPIX(0.07,0.03,0.01) WITH THE STATE-OF-THE-ART METHODS ON DATA OF TRANSPARENT OBJECTS

	MSE*100								**BadPix(0.07)**							
Scene	SPO	ST	SC	OAVC	EPINet	Distg	OACC	ACO	SPO	ST	SC	OAVC	EPINet	Distg	OACC	ACO
Moria	23.64	9.61	3.20	12.27	15.16	15.88	16.42	**0.70**	67.95	51.62	41.36	82.54	85.27	67.90	61.23	**33.27**
Teddy	14.93	13.10	12.39	2.57	11.72	8.44	14.81	**0.27**	18.68	59.07	99.01	63.22	98.62	98.25	86.60	**7.85**
D-90%	4.44	7.28	10.46	**1.06**	20.87	1.94	7.61	1.41	84.47	67.04	94.16	31.41	94.02	81.81	74.61	**29.82**
D-70%	33.27	17.79	12.21	4.15	9.33	10.87	25.59	**1.01**	99.79	67.44	94.82	47.72	94.70	97.19	94.51	**22.46**
D-50%	40.27	18.19	12.38	8.93	17.40	10.71	11.98	**1.94**	99.78	68.16	96.56	59.44	97.19	95.49	94.17	**21.62**
D-30%	45.26	18.72	12.49	14.46	19.89	10.53	10.49	**2.57**	98.27	69.66	95.99	66.80	93.78	96.88	97.59	**23.21**
D-20%	23.31	14.27	11.42	17.30	12.11	6.85	9.21	**5.59**	90.60	74.30	97.09	69.62	90.49	94.38	92.11	**24.96**
Average	26.45	14.13	10.65	8.67	16.64	9.32	13.77	**1.93**	80.07	65.32	88.43	60.15	93.43	90.27	85.83	**23.31**
	BadPix(0.03)								**BadPix(0.01)**							
Moria	79.89	82.33	71.24	89.48	93.22	83.40	78.51	**59.94**	93.92	92.99	90.31	94.94	96.95	94.74	91.19	**81.97**
Teddy	**28.10**	60.96	99.76	84.93	99.41	99.30	91.71	29.67	**34.55**	61.90	99.94	96.61	99.78	99.73	96.62	36.58
D-90%	95.71	74.04	96.88	64.03	94.78	96.61	88.23	**36.65**	98.14	76.85	98.79	96.77	95.11	99.03	94.45	**41.63**
D-70%	99.85	69.69	97.17	73.78	98.99	98.85	97.18	**34.34**	99.96	79.80	98.95	90.13	99.66	99.67	98.89	**40.52**
D-50%	99.87	70.46	97.56	78.76	98.74	98.26	97.86	**33.72**	99.92	71.62	99.10	91.70	99.51	99.36	99.28	**40.44**
D-30%	99.42	71.96	97.74	82.20	97.56	98.43	98.85	**34.54**	99.68	73.12	99.16	93.26	99.17	99.42	99.59	**41.67**
D-20%	96.23	77.71	98.39	83.94	96.26	98.68	97.28	**34.42**	98.42	79.28	99.40	94.21	98.64	99.79	99.11	**40.08**
Average	85.58	72.45	94.39	79.59	96.99	96.21	92.80	**37.61**	89.23	76.50	97.95	93.93	98.40	99.82	97.01	**46.12**

The transparency of the glass itself is gradually decreasing from 90% to 30% by 20% as a step, and the transparency is floating between plus and minus 10% due to random noise and patterns on the glass. To avoid complete opacity, the base transparency of the last set is 20%. "Apple" is a set of images with the same scene, where an apple sits on a smooth marble table. The marble tabletop with different reflectivity reflects the reflection of the apple. This set of data is used to test the robustness of the algorithm to different reflectances. And we use a light field camera Lytro to capture real scene data. All the generated datasets can be downloaded from the following links. https://github.com/HAWKEYE-Group/UrbanLF.

We use the mean squared error (MSE) and bad pixel ratio (0.01,0.03,0.07) to evaluate objective disparity accuracy. For MSE and BadPix, the smaller the better.

$$MSE * 100 = 100 * \frac{1}{N}\sum_{i=1}^{N}(d_{gt}^{(i)} - d_{*}^{(i)})^2,$$
$$BadPix(j) = 100\% * \frac{1}{N}\sum_{i=1}^{N}(|d_{gt}^{(i)} - d_{*}^{(i)}| > j) \quad (17)$$

where N is the sum of pixels. d_{gt}^{i} and d_{*}^{i} are the ground truth value and the estimated result of the ith pixel.

B. Comparisons With State-of-the-Art Methods

In this section, the performance of the proposed method is evaluated using synthetic and real-world data. We compare our method with several top-ranked methods with publications on the 4D Light Field Benchmark, OACC-Net [49], DistgDisp [31], occlusion-aware vote cost(OAVC) [6], EPINet [7], spinning parallelogram operator(SPO) [48], multi-orientation structure tensors(ST) [8], and sparse coding(SC) [9]. In particular, when ACO removes any branch, the cooperation between the two branches disappears. A single ACO branch degenerates into SPO. Thus, the comparison with SPO is chosen in our experiments as a set of ablation experiments to demonstrate the importance of the two-branch structure design.

Since there are adjustable parameters in ACO, we list the three most important optimal parameter selections. The first is α_1, which controls the width of the branches. The second is β_t, which controls how much ACO weakens the foreground object. After the adaptive module is added, these two parameters are adjusted automatically. But they still need an initial default value. The third is t_{tra}, which controls sensitivity to the judgment of transparent regions. We generate a set of training data for different scenes and test the control variables on them. Thus a group of optimal parameters are selected, $\alpha_1 = 0.8$, $\beta_t = 0.8$, $t_{tra} = 0.5$ default.

C. Synthetic Data

We used the class "Transparency" in 4D light field benchmark [11] with test image "Moria" for quantitative evaluation. To test the robustness of the algorithms for different transparency and reflectance, own rendered images *Teddy*, *David*, and *Apple* are also selected for comparison.

1) Evaluation of Transparent Objects: We first evaluated the ability of different algorithms to handle transparent objects. The quantitative results are listed in Tab. I. Fig. 6 shows estimation results and error maps. The proposed ACO achieves state-of-the-art depth accuracy in terms of the average MSE and BadPix scores on both the public dataset and generated data. At the edges of transparent objects, our algorithm achieves good depth continuity. This shows that the transparent objects are clearly separated and do not affect the original information of the background objects. Especially in *Teddy* scene, the boundaries of transparent objects become indistinct in the depth map. This shows that even in the case of similar appearance, ACO can still clearly and effectively separate multi-layer scenes. In addition, deep learning methods perform well when the testing data shares a similar distribution with the train data, but suffer from generalization problems. And depth errors in regions with similar appearance interfere with each other. These methods were not designed with non-Lambertian scenarios in mind. So although they perform well in Lambertian scenes, they still obtain severe artifacts when estimating transparent objects. As aforementioned in Sec. I, this is a limitation of existing depth estimation methods for non-Lambertian depth estimation.

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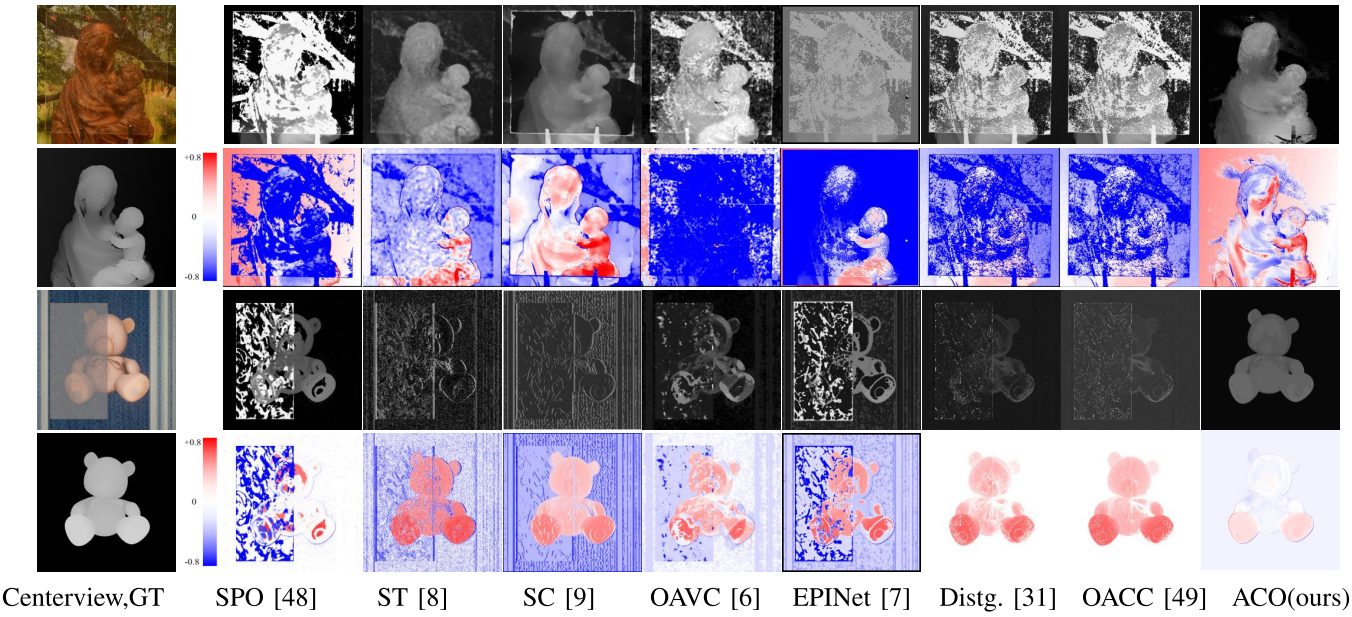


Fig. 6. Visual comparison of estimated depth and error map by different methods on *Moria* of 4D light field benchmark [11] and synthetic data *Teddy*.

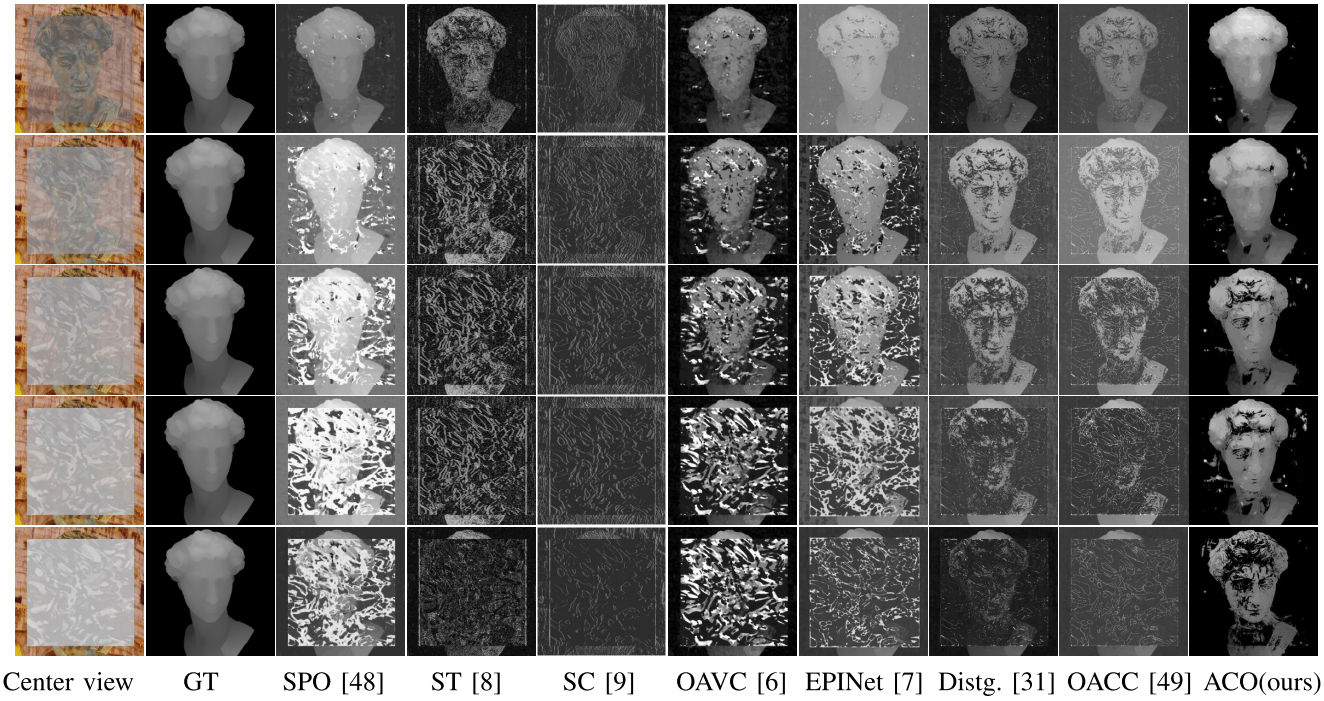


Fig. 7. Visual comparison of estimated depth by different methods on *David* with different transparency.

In addition to the comparison on public datasets, in order to explore the sensitivity of the algorithms to transparency, we conduct a control variable experiment on the data *David* with different transparency. As illustrated in Fig. 7, with the decrease in transparency, the estimation results of the algorithms gradually become inaccurate. This is obvious because less light from background objects is captured by the lens. Different algorithms reach their limits at different transparency. For example, the results of Distg. [31] and OACC [49] erase the original geometry of the statue in the case of 20% transparency. The quantitative results are shown in Tab. I. And the proposed ACO obtains accurate results under different transparency. The strength of our method is that it is not significantly affected by the transparency of non-Lambertian objects and randomly distributed noise due to the proposed adaptive module. The weakness of our method is that it is challenging to reconstruct background objects when the foreground transparency is too low. As shown in *David* scene with transparency of 20%, the result of the depth estimation erases the original geometric structure in detail.

2) Evaluation of Mirror: We then evaluated the ability of different algorithms to handle mirror surfaces. We selected

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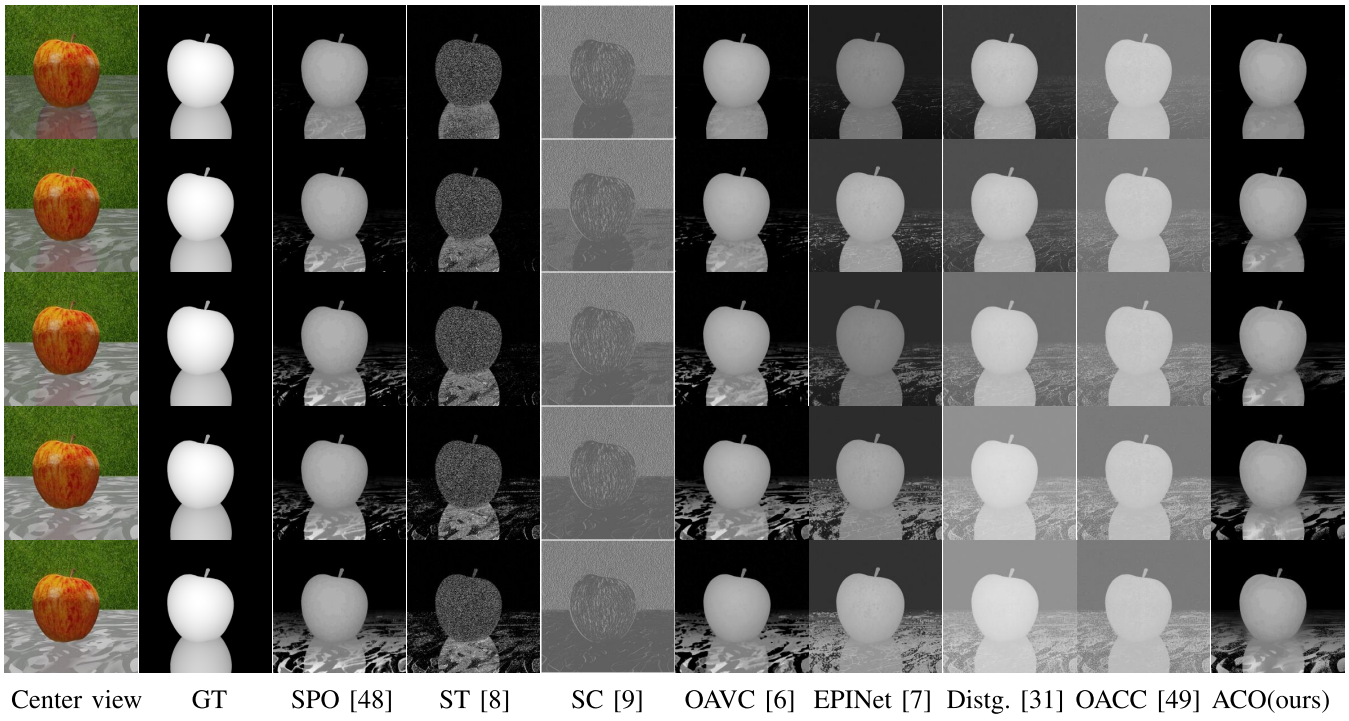


Fig. 8. Visual comparsion of estimated depth by different methods on *Apple* with different reflectivity.

TABLE II
COMPARISON ON MSE AND BADPIX(0.07,0.03,0.01) WITH THE STATE-OF-THE-ART METHODS ON *Apple* SCENE WITH DIFFERENT REFLECTIVITY

	MSE*100								**BadPix(0.07)**							
Scene	SPO	ST	SC	OAVC	EPINet	Distg	OACC	ACO	SPO	ST	SC	OAVC	EPINet	Distg	OACC	ACO
A.-90%	0.04	3.18	9.16	0.04	0.49	0.12	0.26	**0.03**	0.56	19.35	95.36	0.31	4.51	3.16	8.69	**0.27**
A.-70%	0.16	3.21	9.26	0.13	0.65	0.20	0.45	**0.06**	3.57	19.64	95.32	3.31	7.21	5.44	11.49	**1.14**
A.-50%	0.51	3.32	9.53	0.55	1.13	0.42	0.80	**0.23**	8.71	20.48	95.29	9.57	12.18	11.10	16.01	**6.33**
A.-30%	1.30	3.76	9.85	1.33	1.99	0.95	1.34	**0.93**	15.40	23.35	95.13	16.67	19.36	18.62	21.67	**13.85**
A.-20%	1.94	4.11	10.26	1.87	2.67	1.51	1.79	**1.48**	19.79	25.56	95.39	20.99	24.20	22.66	24.95	**19.69**
	BadPix(0.03)								**BadPix(0.01)**							
A.-90%	5.46	26.20	97.95	**2.09**	23.90	18.64	34.66	3.44	27.14	31.21	98.57	**19.37**	48.62	41.62	58.15	24.52
A.-70%	12.62	26.98	97.93	**7.80**	26.03	20.33	37.31	8.72	34.12	31.93	98.56	**25.25**	50.88	43.64	59.16	27.60
A.-50%	18.64	28.07	97.91	15.58	29.34	24.28	40.17	**15.54**	38.92	33.02	98.56	32.43	52.17	45.95	60.10	**31.76**
A.-30%	24.24	31.13	97.92	23.57	33.28	30.16	43.32	**23.05**	43.54	36.09	98.56	**39.11**	52.97	48.10	61.24	40.09
A.-20%	29.09	33.46	97.93	**27.76**	35.47	33.03	45.11	30.65	47.84	38.36	98.56	45.13	53.56	49.14	61.88	**43.11**

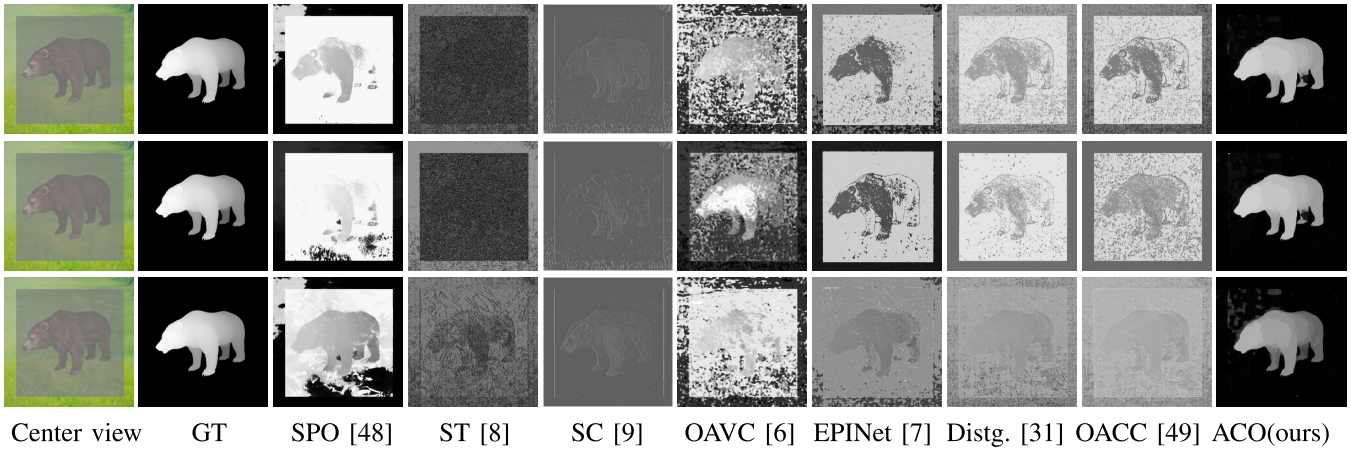


Fig. 9. Visual comparison of estimated depth by different methods on *Bear*. The first row serves as a blank control group. The second line changes the focus plane so that the disparity range is from (-0.62, 0.62) to (-1.05, 0.20). The third row reduces the texture scale and increases the texture density of the non-Lambertian surface.

Apple scene with different reflectivity for evaluation. We still use the idea of controlling variables to gradually reduce the reflectivity of the desk. The set of data *Apple* share the same scene except for the desktop with different reflectivity,

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TABLE III

COMPARISON ON MSE AND BADPIX(0.07,0.03,0.01) WITH THE SATE-OF-THE-ART METHODS ON *Bear* SCENE WITH DIFFERENT NOISE.(RED = UP, AND BLUE = DOWN)

	MSE*100								**BadPix(0.07)**							
Scene	SPO	ST	SC	OAVC	EPINet	Distg	OACC	ACO	SPO	ST	SC	OAVC	EPINet	Distg	OACC	ACO
B.blank	23.70	8.53	23.68	12.74	83.74	72.57	77.33	**0.20**	64.29	63.03	93.06	88.51	81.08	88.11	88.40	**9.00**
B.focus	13.62 ↓10.08	8.08 ↓0.45	23.35 ↓0.33	3.28 ↓9.46	80.12 ↓3.62	80.84 ↑3.62	63.86 ↓13.74	**0.19** ↓0.1	67.56 ↑3.62	64.86 ↑1.83	92.60 ↓0.46	70.58 ↓17.93	82.37 ↑1.29	91.25 ↑3.14	92.13 ↑3.73	**8.88** ↓0.12
B.texture	17.16 ↓6.54	9.07 ↑0.54	23.17 ↓0.51	18.72 ↑5.98	27.84 ↓55.9	23.43 ↓49.14	25.89 ↓51.44	**0.39** ↑0.19	49.77 ↓14.52	78.65 ↑15.62	93.57 ↑0.51	92.93 ↑4.42	76.49 ↓4.59	83.04 ↑5.07	83.28 ↓5.12	**23.70** ↑14.7
	BadPix(0.03)								**BadPix(0.01)**							
B.blank	68.91	64.10	96.61	94.57	82.78	91.08	92.58	**21.20**	75.39	64.86	98.26	97.90	87.00	95.61	97.29	**24.16**
B.focus	79.44 ↑10.53	65.76 ↑1.66	96.36 ↓0.25	91.17 ↓3.40	87.22 ↑4.44	97.43 ↑6.35	94.44 ↑1.86	**23.78** ↑2.58	88.29 ↑12.9	66.48 ↑1.62	98.17 ↓0.09	96.15 ↓1.75	92.08 ↑5.08	99.00 ↑3.39	97.19 ↓0.1	**26.80** ↑2.64
B.texture	59.29 ↓9.62	81.02 ↑16.92	96.94 ↑10.08	97.16 ↑2.59	81.10 ↓1.68	88.28 ↓2.80	90.63 ↓1.95	**24.61** ↑3.41	70.02 ↓5.37	82.81 ↑17.95	98.46 ↑0.20	98.90 ↑1.00	88.87 ↑1.87	92.89 ↓2.72	96.95 ↓0.34	**24.64** ↑0.48

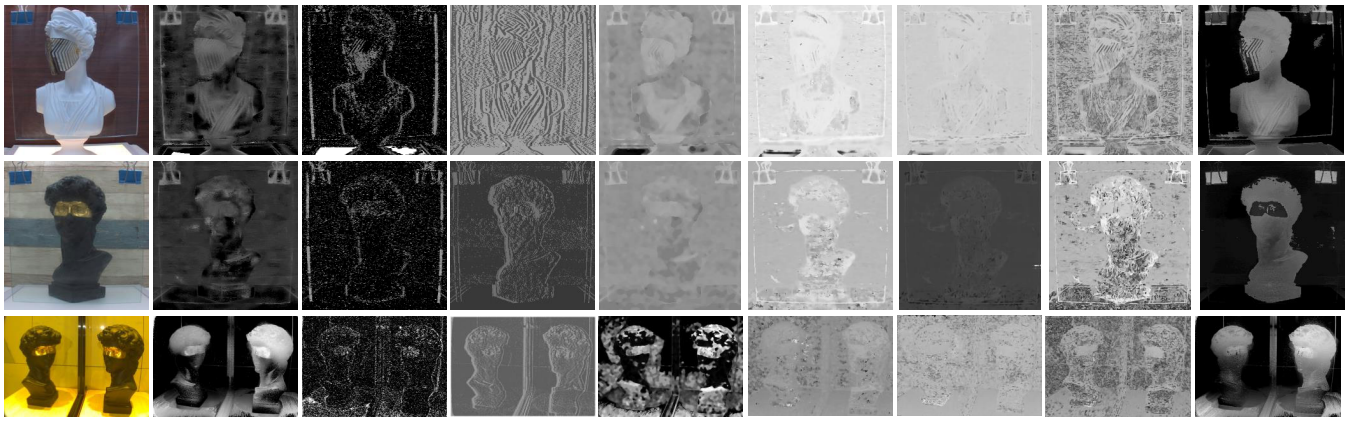


Fig. 10. Visual comparsion of estimated depth by different methods on real-world images.

which decreases sequentially from 90% to 20% in steps of 20%. The reflection of the apple on the desktop becomes decreasingly clear, which brings about the difficulty in depth estimation. As shown in Fig. 8, the performance of the algorithms gradually decreases with the decreasing reflectivity. The low reflectivity results in the mirror surface exhibiting stronger Lambertian features, which leads to the underestimation of depth for large areas. Tab. II shows that ACO has good performance under different reflectivity. The adaptation module of ACO promotes its insensitivity to the coefficient of reflection.

3) Evaluation of Noise: In our experiments, we find that the depth estimation of non-Lambertian objects is not only affected by the reflectance coefficient but also fluctuated by the disparity range of scenes and the texture of non-Lambertian surfaces. We designed experiments to demonstrate the influence of these factors on the algorithms. The set of data *Bear* consists of three LF images. The first LF served as a blank control group. On the basis of the first LF, the second changes the position of the focus plane, which makes the disparities of the background and transparent object have different positive and negative properties. The third LF changes the scale and denseness of the texture on the non-Lambertian surface based on the first LF. As shown in Fig. 9, these factors have different effects on different algorithms. For example, after changing the position of the focus plane, OAVC [6] is able to better separate the information of the foreground and background with different positive and negative properties. ST [8] achieves higher accuracy when dealing with transparent objects with small-scale textures, due to the structure tensor being better at describing local features. The quantitative experimental results in Tab. III show that ACO is not sensitive to these influencing factors and the MSE is kept at a low range.

D. Real-World Data

We also conducted experiments on real-world scenes by comparing our method with SPO [48], ST [8], SC [9], OAVC [6], EPINet [7], Distg. [31], and OACC [49]. Real-world images are captured with a Lytro Illum camera in this study. Fig. 10 compares the qualitative results. The noises of real images are more serious, and the lighting condition is more complicated than that of synthetic images. This presents a challenge to the robustness of the algorithms. ACO still performs well visually.

VI. CONCLUSION

In this study, Adaptive Cross Operator has been proposed to solve the problem of light field depth estimation for non-Lambertian objects represented by transparent objects and mirror surfaces. The experimental results show that ACO has good performance under different transparency and reflection coefficients. Firstly, it is discovered that the unique photo inconsistency in the non-Lambertian regions on EPIs

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is the main cause of the depth estimation error. Based on this, ACO is designed with a double-branch structure to avoid interference from photo inconsistency. Identifying the bimodal characteristics of ACO response value function, non-Lambertian regions can be marked out. Then, we implement an ACO-based depth estimation algorithm compatible with non-Lambertian light fields to address the limitations of existing algorithms. The adaptive module of the ACO helps it to obtain accurate estimation results for scenes with different reflection coefficients.

Due to the inter-branch cooperation strategy of ACO, its application scope is not limited to X-shaped linear structures on EPI. We consider that ACO has the potential to deal with more complex reflection scenes as well, i.e. the presence of both reflection and refraction on the surface of an object.

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